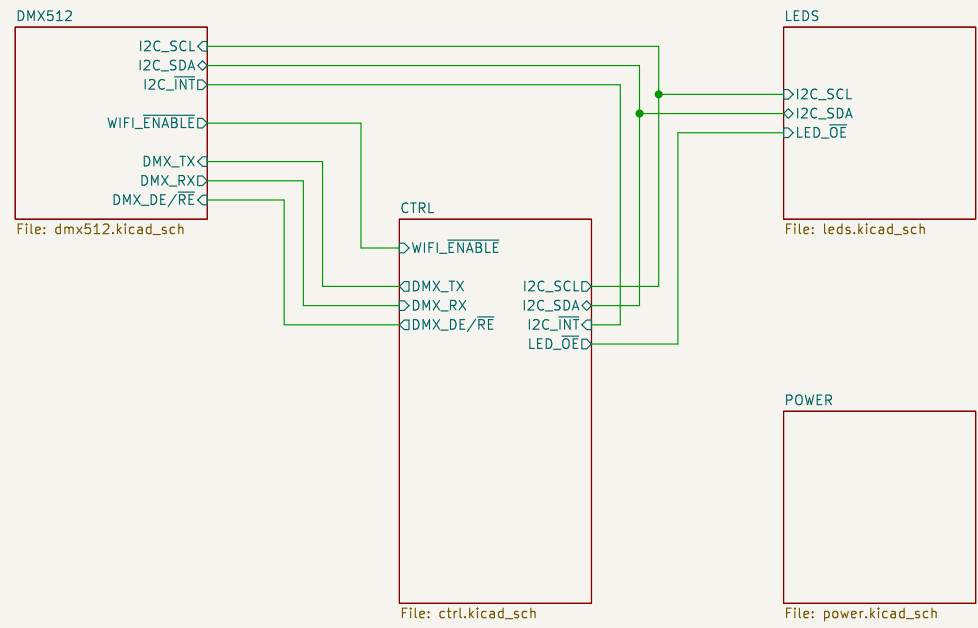


LED Beacon



- H1 MountingHole
- H2 MountingHole
- H3 MountingHole
- H4 MountingHole

Wires & Bits		
Sheet: /		
File: emf_beacon.kicad_sch		
Title:		
Size: A4	Date: 2026-04-10	Rev: 0
KiCad E.D.A. 9.0.9		Id: 1/5

Controller

USB Programming

VCC Monitoring

Temperature Monitoring

Auto reset

Status LED

Does this need to be a switch?
Cheaper switch?
Side actuation?

Wires & Bits

Sheet: /CTRL/
File: ctrl.kicad_sch

Title:

Size: A4 Date: 2026-04-10 Rev: 0
KiCad E.D.A. 9.0.9 Id: 2/5

USB_C_Receptacle_USB2.0_16P

VBUS A4

CC1 A5

CC2 B5

D- A7

D+ B7

D- A6

D+ B6

SBU1 A8

SBU2 B8

SHIELD

GND

R1 5k1

R2 5k1

GNND

D5 SS34

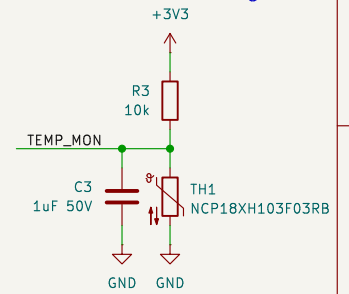
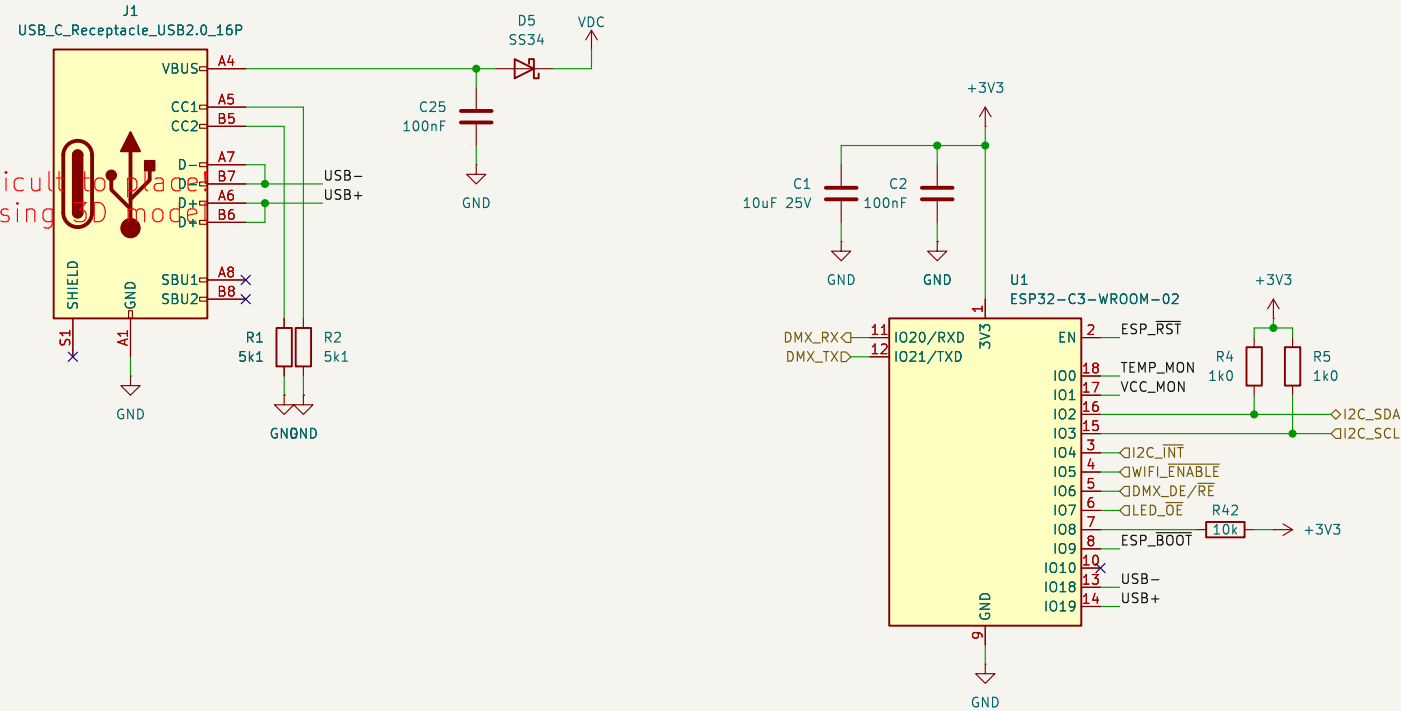
C25 100nF

GND

5VDC

USB-

USB+



Auto reset

D

+3V3

R41
10k

ESP_RST

D6
BAT54WS

C26
100nF 10uF

C27
25V

GND

GND

GND

Short press resets the ESP32
Long press resets the ESP32
and puts it in bootloader mode

Auto reset

D

+3V3

R41
10k

ESP_RST

D6
BAT54WS

C26
100nF 10uF

C27
25V

GND

GND

GND

Short press resets the ESP32
Long press resets the ESP32
and puts it in bootloader mode

Auto reset

D

+3V3

R41
10k

ESP_RST

D6
BAT54WS

C26
100nF 10uF

C27
25V

GND

GND

GND

Short press resets the ESP32
Long press resets the ESP32
and puts it in bootloader mode

Status LEDs?

Sheet: /CTRL/		D
File: ctrl.kicad_sch		
Title:		
Size: A4	Date: 2026-04-10	Rev: 0
KiCad E.D.A. 9.0.9		Id: 2/5

DMX512

To XLRs

J2 Conn_01x06

J3 Conn_01x06

R8 1M0

C5 1nF 2kV

GND_ISO

DMX_DATA-

DMX_DATA+

DMX_AUX-

DMX_AUX+

DMX_DATA-

DMX_DATA+

DMX_AUX-

DMX_AUX+

R10 4.7k

R11 4.7k

GND_ISO

D1 SM712

GND_ISO

Failsafe Bias

+3V3_ISO

C7 10uF 25V

C8 100nF

GND_ISO

GND_ISO

DMX_DATA+

DMX_DATA-

Isolated RS485 TRX

+3V3_ISO

+3V3

U3 CA-IS3092W

C9 100nF

C10 10uF 25V

GND

GND

OUT OF STOCK!

DMX_TX

DMX_DE/RE

DMX_RX

R43 310k

+3V3

GND

GND_ISO

GND_ISO

GND

Make a symbol for CA-IS3092W CA-IS3092VW

GPA7 & GPB2:7 for?

SW1 SW_DIP_x10

DMX_ADDR0

DMX_ADDR1

DMX_ADDR2

DMX_ADDR3

DMX_ADDR4

DMX_ADDR5

DMX_ADDR6

DMX_ADDR7

DMX_ADDR8

WIFI_EN via GPIO expander?

GND

MCP23017T-E/SS

U2

+3V3

C6 100nF

GND

SDA

SCL

R12 10k

+3V3

I2C_INT

R13 10k

+3V3

INTB

INTA

RESET

A2

A1

A0

GND

GND

I2C Addr: 0x20

Wires & Bits

Sheet: /DMX512/

File: dmx512.kicad_sch

Title:

Size: A4

Date: 2026-04-10

KiCad E.D.A. 9.0.9

Rev: 0

Id: 3/5

https://dfimg.drobot.com/wiki/23269/DFR0944_esp32-s3-based-dmx512-smart-lighting-controller_schematics_V1.0.pdf

<https://www.ti.com/lit/an/slla272d/slla272d.pdf>

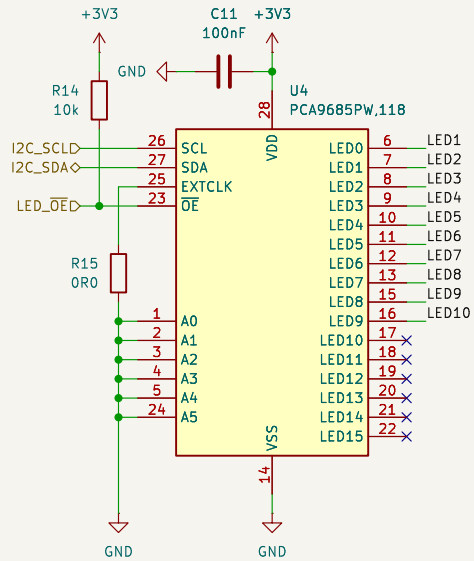
KiCad E.D.A. 9.0.9

LED Driver

LED Enable

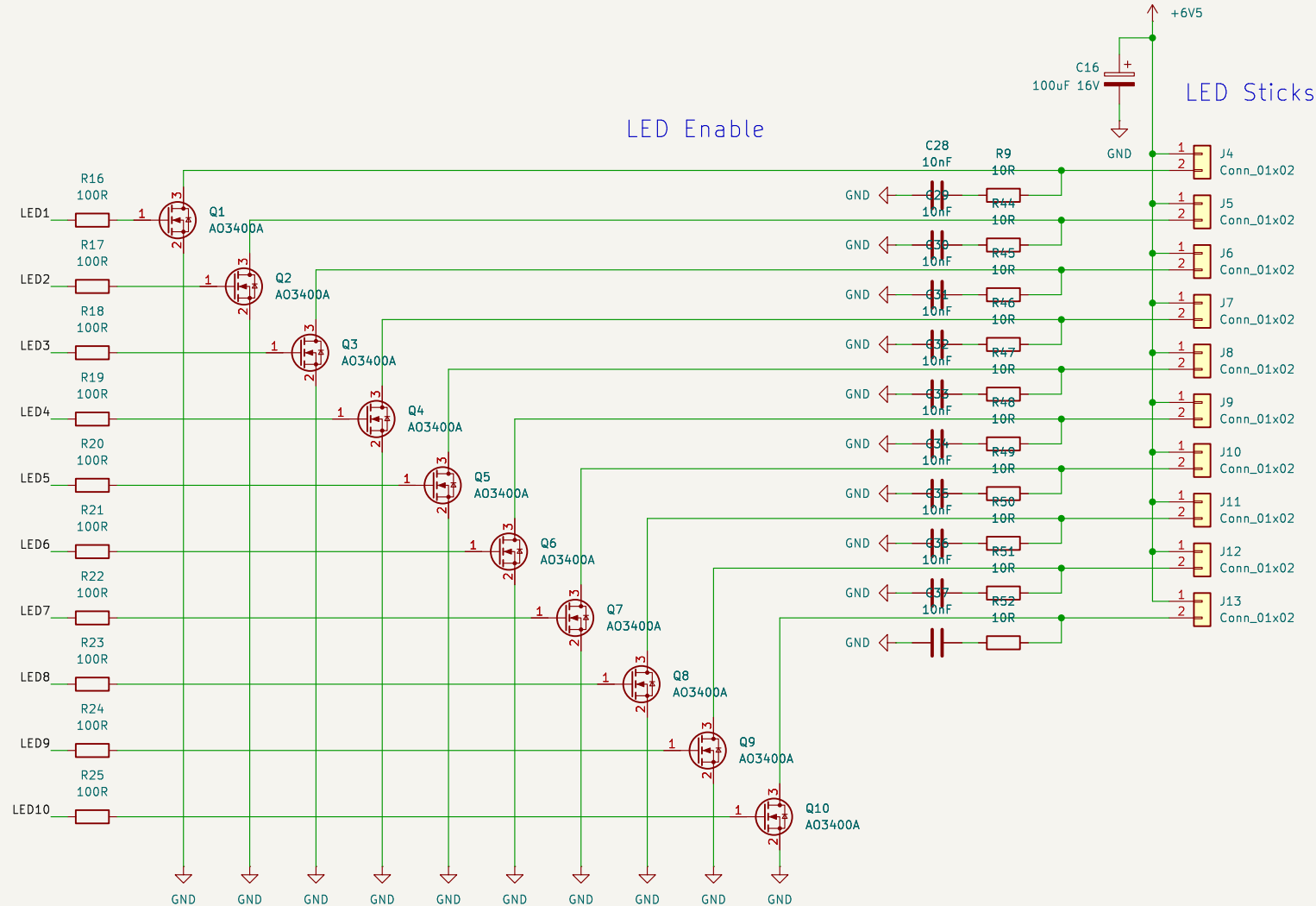
LED Sticks

I2C → PWM



I2C Addr: 0x40

A string of 6x LEDs is 6.4Vf @ 0.3A



Wires & Bits

Sheet: /LEDS/
File: leds.kicad_sch

Title:

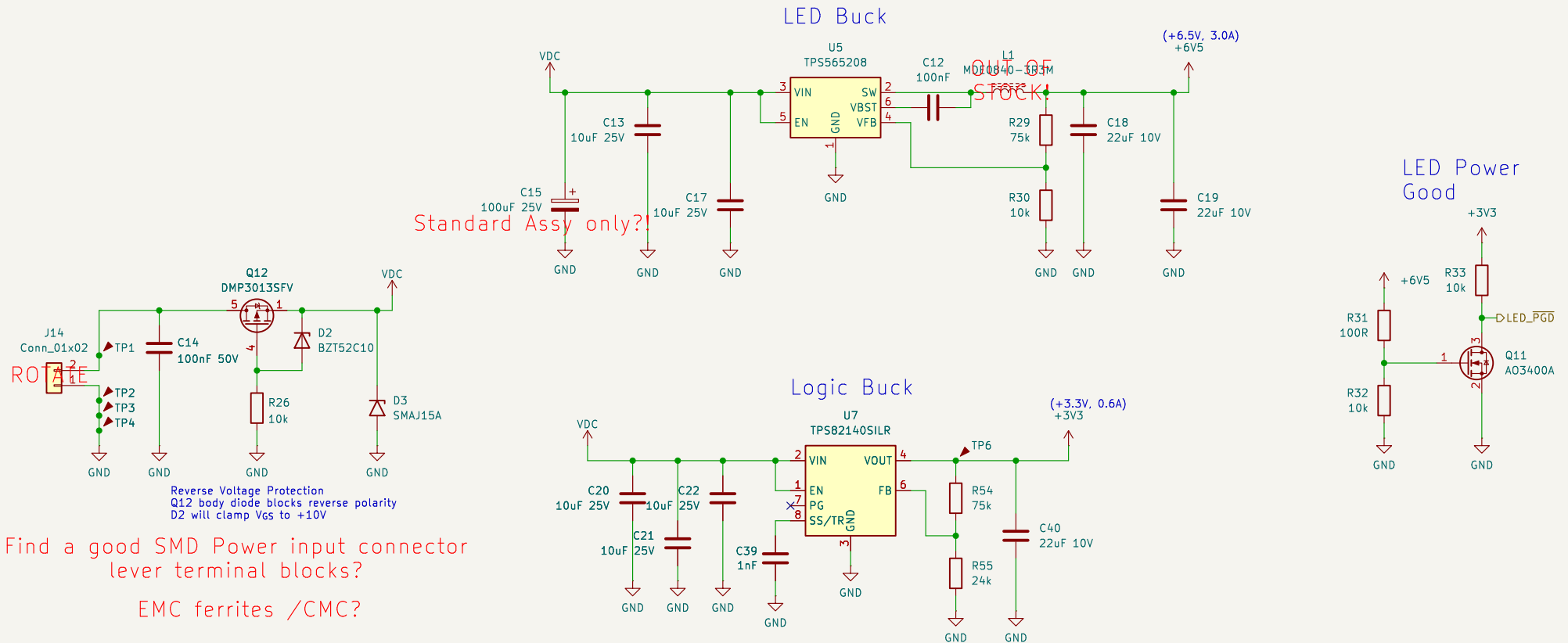
Size: A4
KiCad E.D.A. 9.0.9

Date: 2026-04-10

Rev: 0

Id: 4/5

Power Supplies



Wires & Bits

Sheet: /POWER/
File: power.kicad_sch

Title:

Size: A4

Date: 2026-04-10

KiCad E.D.A. 9.0.9

Rev: 0

Id: 5/5